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A Deep Learning Approach for Identifying Endangered Animals in Bangladesh

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***Abstract*—Preserving endangered animals from extinction is an overwhelming challenge for researchers throughout the globe. Monitoring wildlife in their natural habitat is crucial. This proposed work develops an automated system that can detect endangered animals in nature. It might be challenging to manually identify endangered animals since there are so many distinct species. For that, we constructed a custom dataset carefully consisting of 1,880 images representing 10 endangered animal species, sourced from multiple online platforms. The images performed preprocessing, including resizing, normalization, and data augmentation techniques such as random shear, zooming, and horizontal flipping to enhance model generalization. We assessed five deep learning(DL) models on this dataset: Custom Convolutional Neural Network (CNN), VGG16, InceptionV3, MobileNetV2, and DenseNet121. DenseNet121 showed the best results, achieving 99.67% accuracy on both the training and testing datasets, which means it generalizes better, learns faster, and avoids overfitting compared to the other models. Furthermore, to improve model transparency and interpretability, we used Explainable Artificial Intelligence(XAI) approaches, notably LIME, which highlighted the key image regions that contribute to the model's predictions. These findings demonstrate how well DenseNet121 can identify endangered animal species, offering a dependable tool to aid in conservation and wildlife monitoring.**

***Index Terms*—Endangered species identification, Computer vision, Convolutional Neural Network, Deep Learning.**

I. Introduction

The complexity and profusion of life on Earth are collectively referred to as biodiversity. The reason for this is the most complicated and critical aspect of our planet. Biological diversity is crucial both for ecological and economic purposes. Precise identification abilities are required for several activities, including the study of a region's biodiversity richness, the monitoring of imperiled species populations, the assessment of the influence of the changing climate on the spread of species, and the implementation of vegetation control measures.

Currently, advanced technologies such as machine learning (ML) and DL can train computers to recognize a diverse array of objects from images. Visual information provides irrefutable evidence of an animal's presence by capturing image frames and identifying them as threatened species without human intervention or support, according to wildlife researchers. This would assist researchers, ecologists, and scientists in classifying and preserving these animals. CNNs utilize relatively small amounts of preprocessing in comparison with various other image classification techniques. This freedom from human intervention in feature design and past knowledge is a significant advantage in CNN. They have a wide range of applications in medical image processing, image classification, recommendation systems, and image and video recognition.

In this study, our primary objective is to utilize several deep-learning techniques to detect endangered species from their images automatically. This automated animal identification system can also assist in the prevention of animal-vehicle collisions, which can lead to property damage, injury, and fatality.

II. Literate Review

This research presents a modular and interoperable system for the real-time identification of rhinoceroses and vehicles, shown by a field demonstration at Knowsley Safari in the UK. A custom model is trained using 350 photos of vehicles and 350 photographs of rhinos by transfer learning using the Faster RCNN Resnet 101. The Real-Time Messaging Protocol (RMTP) is used to send video broadcasts from a DJI Mavic Pro 2 drone. The optimally trained Faster RCNN model attained a mAP of 0.83 at IOU 0.50 and 0.69 at IOU 0.75, respectively [1]. Another study proposes a highway animal detection system and driver warning architecture. The study extracts picture regions and uses ML algorithms to identify them as either animal or non-animal. For that, five methods were utilised for comparing two machine-learning algorithms by traversing synthetic picture pixel structures. The results show that K-Nearest Neighbours (KNN) surpasses Random Forest (RF) for accurate roadside animal recognition [2]. One more study demonstrated the use of CNN to identify wildlife. 3.2 million animal photos were utilized. Preventing wildlife poaching, resolving human-animal conflicts, and accurately classifying animals (with an accuracy of 86%) are all possible through this system, which also shows the picture of the identified animal [3]. This technique is effective in identifying endangered species with diverse sizes, shapes, textures, colours, and other characteristics. Using a customised endangered wildlife dataset containing 12 classifications and multiple DL models, including Wildect-YOLO, endangered species are detected in real-time. The CSPDarkernet53 backbone is made up of DenseNet blocks and a residual block

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that keeps important feature information safe and allows for strong and accurate deep spatial feature extraction. This model exceeds state-of-the-art models with an mAP of 96.89%, an F1 score of 97.87%, and precision with a value of 97.18% at 59.20 FPS. [4].

To determine which animals were in a given set of millions of photographs, this research used convolutional networks, a kind of ML. This study outlines key DL techniques for automated animal identification, segmentation, and recognition, and provides a brief analysis and comparison of various approaches [5]. The goal of this effort is to build an autonomous object recognition system that uses an IP camera-based smartphone to identify insect pests in digital photos and videos, with the aim of reducing their impact on crops, using a custom dataset called IP-23. This technique uses YOLOv5 (n, s, m, l, and x), YOLOv3, YOLO-Lite, and YOLOR object detection designs. Eight models are compared using performance indicators. On the IP-23 dataset, the newly developed YOLOv5x model surpasses the state-of-the-art model by achieving a 96% F1-score, 94.5% accuracy, 97.8% recall, and 98.3% average precision (mAP@0.5:0.95) [6]. Using a tailored endangered animal dataset and few-shot learning in several scenarios with geographical dependence, this technique proposes reacting to an ever-changing environment for endangered species identification. The spatial-constraints model can stop the mixing of foreground and background species because the geographic backgrounds don't match. It does this by having parts with a knowledge dictionary and a relation network [7]. This effort focuses on animal species identification to reduce human-vehicle accidents in rural and highway regions. Using a deformable CNN and four R-CNN models in three popular animal datasets to increase species identification speed and accuracy. In real-time wildlife identification, Deformable Mask R-CNN surpasses all other models, achieving 98.4% accuracy on Snapshot Serengeti, 93.3% on BCMOTI, and 97.6% on Snapshot Wisconsin [8]. This experiment used transfer learning to recognize Malayan civets, Visayan leopard cats, spotted deer, and warty pigs in Negros Island woods. They created a web app to capture and timestamped camera data. A solar power bank powered this setup, which included the Raspberry Pi 3B+ and camera module. YOLOv5n, a lightweight object identification technique, successfully detected all four species. The trained model had 91% mAP, 64% mAP@0.5:0.95, and average precisions of 94% for Visayan warty pig, 88% for Visayan leopard cat, 91% for Malayan civet, and 91% for Visayan spotted deer using 4.5 GFLOPs [9].

One more study investigates animal detection and identification techniques used in images, including CNN, YOLO, SSD, and MASK RCNN. Following the survey, VGGNET emerged as the most precise method for animal identification, achieving 96.6% accuracy. For object detection, SSD achieved 0.67 APn and YOLO 0.55 APn, but their combination resulted in a higher APn of 0.73 [10]. This paper introduces a DL approach that trains a CNN using web-retrieved animal images. This approach can identify sparsely scattered animals in several aerial photos. With an acceptable detection threshold, they identified 95% of known polar bear photos in aerial survey shots and 99.4% of validation images with just background features [11]. Another study introduces a DL model that uses CNN to identify endangered species in camera trap data with an accuracy rate of 94.6% [12]. Another study [13], classified endangered wildlife in mining-affected areas using DL. With an average accuracy and F1-score above 0.97, the model classified photos of various quality and noise levels. One different investigation unveils EndanSys, that makes use of a ResNet50 model that was trained on 4,093 photos of animals in Colombia. The model classified endangered species with an accuracy of 89.23% [14].

III. Research Methodology

This part covers the data collection and training procedure for the endangered animal detection model. Training parameters and associated techniques are also covered here. Finally, we assessed the performance of the trained model as well as the evaluation metrics.

A. Data Collection

We created the dataset by gathering photos from different online platforms. All of our authors double-checked each image using visual traits like body shape, patterns, textures, and unique traits that corresponded with species descriptions to make sure the labelling was accurate. The dataset contains 10 endangered animal classes such as 'Cuon Alpinus', 'Banteng', 'Hog Deer', 'Tiger', 'Pangolin', 'Leaf Monkey', 'Hispid Hare', 'Hoolock hoolock', 'Elephas Maximus', and 'Wild Water Buffalo'.

Fig. 1: Dataset snippet

The collection comprises a total of 1880 endangered animal photos. All images were resized to 150×150×3 to align with our network's dimensions. An illustration of several endangered animal types is shown in Figure 1, which also includes fourteen random photos taken from the animal dataset that was developed.

A statistical overview of the developed dataset is mentioned below, which also includes a breakdown of the exact number of endangered animal images in each category. The entire collection of endangered species images is distributed over ten distinct categories. These categories include: ”Elephas Maximus” with 428 images, ”Hoolock hoolock” with 391 images, ”Cuon Alpinus” with 312 images, ”Hispid Hare” with 202 images, ”Hog Deer” with 101 images, ”Tiger” with 99 images, ”Pangolin” with 99 images, ”Leaf Monkey” with 89 images, ”Wild Water Buffalo” with 88 images, and ”Banteng” with 71 examples. For our research, we randomly assigned 80% of the collected images to be used for training, 10% will be used for testing, and 5% will be used for validation.

B. Feature Extraction

Techniques for feature extraction are used to locate and extract pertinent characteristics from a picture.

C. Image Resizing

Image scaling methods help to change an image's scale. Resizing may be done to alter an image's aspect ratio or shrink or enlarge it.

D. Image Augmentation

Data augmentation uses rotation, flipping, and zooming to alter pictures. Diversifying the dataset improves the model's generalization and performance on new data.

E. Binarization

To create a binary image, it replaces all values higher than a globally defined threshold with 1 and all values lower than it with 0.

F. Convolutional Neural Network (CNN)

Convolutional Neural Network is a popular DL architecture that extracts features from visual data for classification, detection, and identification. CNNs are made up of several layers, including convolutional layers, pooling layers, and fully connected layers, which allow the network to learn hierarchical patterns from raw pixel data.

This work presents a custom CNN consisting of three convolutional layers with progressively increasing filters (32, 64, 128), followed by MaxPooling layers, a Flatten operation, and two fully connected layers (512 neurones and a concluding Softmax layer for multi-class classification).

G. VGG16

VGG16 is a well-known deep CNN design that stands out for its simplicity and uniformity. This study employed the VGG16 model, pre-trained on the ImageNet dataset, as a fixed feature extractor by eliminating its upper classification layers. The input images were resized to 150×150×3 to align with the modified architecture. A Global Average Pooling layer was added, succeeded by a fully connected Dense layer comprising 512 units with ReLU activation. A Softmax layer was incorporated to facilitate multi-class classification aligned with the number of endangered species categories. In the training process, the pre-trained convolutional layers were kept static to utilise existing features and minimise computational expense, with only the newly introduced layers undergoing updates. The model underwent optimisation via the Adam optimiser and was trained utilising categorical cross-entropy loss.

H. InceptionV3

InceptionV3 is a popular CNN design that has been optimized for great accuracy while requiring few processing resources. This research makes use of the InceptionV3 model, with the exception of the top classification layers, which have their weights pre-trained on the ImageNet dataset. The custom dataset is used to set the input size, which is 150×150×3. A Global Average Pooling layer, a Dense layer with 512 ReLU-activated units, and a Dense Softmax layer proportional to the number of classes are added to the model to make it suitable for the endangered species classification task. During training, the base model layers are frozen to preserve learnt feature representations, while only the newly added layers are trained. To ensure efficient multi-class classification, the model is assembled using the Adam optimizer with categorical cross-entropy as the loss function.

I. MobileNetV2

MobileNetV2 is a streamlined deep CNN, designed for optimal efficiency on mobile and embedded devices. It utilises depthwise separable convolutions in conjunction with inverted residual architectures and linear bottlenecks to markedly decrease computational cost while maintaining accuracy. This study employs MobileNetV2 with pre-trained ImageNet weights, omitting the top classification layers. The model uses 150×150×3 input images, a Global Average Pooling layer, a Dense layer with 512 ReLU-activated units, and softmax activation for multi-class classification. Throughout the training process, the foundational layers remain static, while only the newly incorporated layers undergo modification. The model is put together using the Adam optimiser, and the loss function is categorical cross-entropy.

J. DenseNet121

An architecture for deep CNN called DenseNet121 is based on dense blocks rather than residual connections. This technique allows for feature reuse across layers and improves parameter efficiency. DenseNet121 has four dense blocks with several convolutional layers, including a transition block that compresses output feature maps. Dense blocks have several levels that are tightly connected to all other layers; thus, information may travel faster.

For our research, at first the images are shrunk to 150×150 pixels, and their pixel values are adjusted to fall within the range of 0 to 1. After that, we just utilize a DenseNet121 model for feature extraction, which was pretrained on ImageNet but without the top classification layer. A GlobalAveragePooling2D layer combines the output feature maps into a single vector. The vector then goes through a Dense layer with

512 ReLU-activated units (optionally with L2 regularization) before class probabilities are produced by a final Dense layer with 10 softmax-activated units. The newly added 512-unit and 10-unit layers are the only ones that train because all of the DenseNet121 levels are frozen at the start. We used the Adam optimizer to train the model, and our learning rate was 0.001. Accuracy is measured using categorical cross-entropy, which is also used as the loss function. Data is enhanced in real-time using techniques such as random shear, zoom, and horizontal flipping. Training is done in batches of 32 for 50 epochs, using an 80% training, 5% validation, and 15% test split.

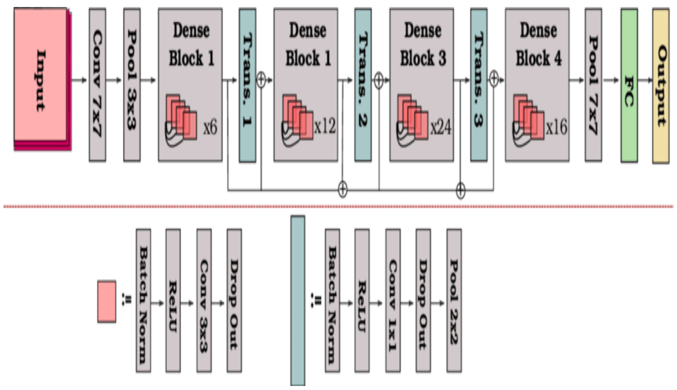

Fig. 2: DenseNet121 model architecture.

TABLE I: Architectural comparison of applied models.

Aspect	Custom CNN	VGG16	InceptionV3	MobileNetV2	DenseNet121
Input Size	150×150×3	150×150×3	150×150×3	150×150×3	150×150×3
Convolution Layers	3 Conv2D (32, 64, 128 filters)	Pre-trained layers with 3×3 convolution	Pre-trained layers with factorized convolution	Pre-trained layers with depthwise separable convolution	Pre-trained layers with densely connected convolution
Pooling	MaxPooling (2D)	Global Average Pooling	Global Average Pooling	Global Average Pooling	Global Average Pooling
Fully Connected Layers	Dense(512, ReLU), Dense (Softmax for classes)	Dense(512, ReLU), Dense (Softmax for classes)	Dense(512, ReLU), Dense (Softmax for classes)	Dense(512, ReLU), Dense (Softmax for classes)	Dense(512, ReLU), Dense (Softmax for classes)
Trainable Layers	All layers	Added dense layers only	Added dense layers only	Added dense layers only	Added dense layers only
Optimizer	Adam	Adam	Adam	Adam	Adam
Loss Function	Categorical Crossentropy	Categorical Crossen-tropy	Categorical Crossentropy	Categorical Crossentropy	Categorical Crossentropy

A comparative overview of five DL architectures is shown in Table I. These models include Custom CNN, VGG16, InceptionV3, MobileNetV2, and DenseNet121. The models are compared according to depth, convolution technique, pooling method, activation function, and input size. Among them, DenseNet121 employs dense connections and has 121 layers, making it the deepest; on the other hand, MobileNetV2 prioritizes lightweight architecture with depthwise separable convolutions.

Figure 3 displays the workflow diagram we followed. To develop our automated endangered animal detection system, we conducted research on several deep-learning classifiers, including MobileNetV2, InceptionV3, VGG16, Lightweight CNN, and DenseNet121.

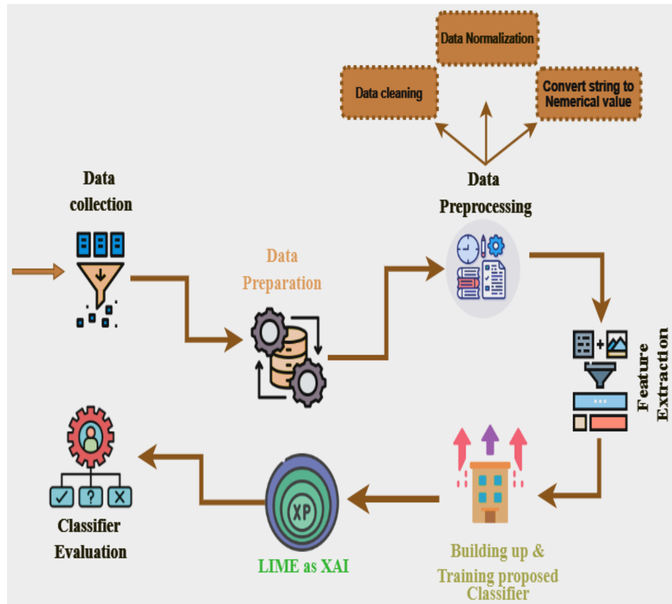

Fig. 3: Work-flow diagram.

IV. Result Analysis and Discussion

This paper presents a DL-based approach for detecting endangered animal species in Bangladesh through image classification. The authors developed a custom dataset of 1,880 images covering 10 endangered species and applied preprocessing techniques such as resizing, normalization, and data augmentation. The evaluation of different models revealed varying levels of computational efficiency. VGG16 required the most resources, with a training time of 8067s and memory usage of 2107MB. Lightweight CNN (3524s, 1137 MB) and InceptionV3 (2874s, 1103 MB) demonstrated moderate efficiency, while MobileNetV2 (1754s, 987MB) and DenseNet121 (1683s, 1003MB) achieved the fastest training times with minimal memory consumption. All experiments were conducted on Google Colab with a Core i5 processor and 8 GB of RAM.

A. Experimental Results

The training and testing accuracy of our utilized model is shown in Table II. In this case, we looked at all models across 50 epochs to determine the optimal epoch number that would provide the highest training accuracy.

TABLE II: Performance comparison of applied models.

Model	Training Accuracy	Testing Accuracy	No of Epochs	Best Epoch
VGG16	93.46%	95.37%	50	24
Custom CNN	98.81%	98.56%	50	37
InceptionV3	99.32%	98.51%	50	15
MobileNetV2	99.87%	99.40%	50	17
DenseNet121	**99.67%**	**99.67%**	**50**	**19**

From the above accuracy in Table II, it is clear that the DenseNet121 generates the highest accuracy among the other five applied models.

Table II presents a comparison of the performance metrics for five DL models: VGG16, Custom CNN, InceptionV3, MobileNetV2, and DenseNet121, all of which were trained on the endangered-species image dataset. Among these models, DenseNet121 exhibits the highest accuracy, attaining 99.67% on both training and testing datasets, with peak performance achieved at epoch 19. MobileNetV2 exhibits a test accuracy of 99.40%, achieved at epoch 17, while displaying minor indications of overfitting, as evidenced by its higher training accuracy of 99.87%. InceptionV3 demonstrates strong performance, achieving a testing accuracy of 98.51% at an early convergence point (epoch 15), although a minor gap between training and testing accuracy suggests possible overfitting. The Custom CNN attains a test accuracy of 98.56%, yet necessitates a greater number of epochs (37) for convergence, suggesting slower training dynamics. Finally, VGG16 demonstrates the lowest performance, achieving only 95.37% test accuracy at epoch 24, which highlights its limitations in addressing this fine-grained classification task. DenseNet121 demonstrates an optimal balance of high accuracy due to its dense connectivity pattern, which ensures that each layer receives inputs from all antecedent layers. This design improves gradient flow, encourages feature reuse, and reduces the disappearing gradient issue. This means that DenseNet121 is great for fine-grained classification tasks like endangered species identification since it can extract richer and more discriminative features with fewer parameters.

TABLE III: Classification report of applied model.

Model	Precision	Recall	F1-Score	Support
VGG16	0.97	0.93	0.95	1880
Custom CNN	0.99	0.99	0.99	1880
InceptionV3	0.99	0.99	0.99	1880
MobileNetV2	0.99	0.99	0.99	1880
DenseNet121	**1.00**	**1.00**	**1.00**	**1880**

Table III presents a detailed performance evaluation of the five DL models utilizing precision, recall, and F1-score on 1,880 samples. DenseNet121 achieved 1.00 F1-scores, precision, recall, and balanced predictive capacity with perfect classification performance. After MobileNetV2, InceptionV3, and the Lightweight CNN, they all attain almost perfect metrics (0.99 recall, precision, and F1-score), indicating great classification accuracy with few misclassifications. VGG16 performs poorly on accurate classification problems with closely related classes, with a recall of 0.93, a precision of 0.97, and an F1-score of 0.95. These tests demonstrate that DenseNet121 consistently beats all other networks in recognizing endangered species photographs in the dataset.

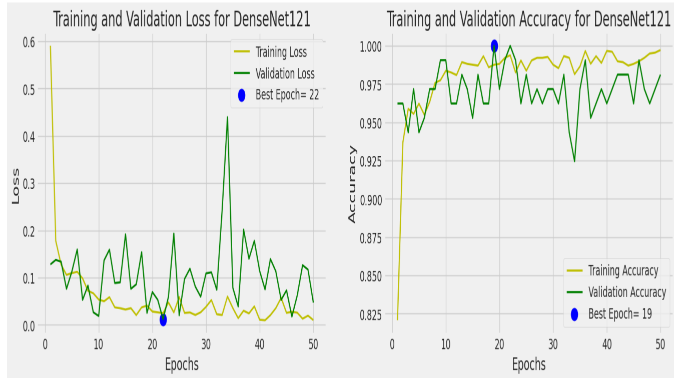

Fig. 4: Testing and validation accuracy of DenseNet121 model.

Figure 4 demonstrated the DenseNet121 model's 50-epoch training progress. The left plot depicts training and validation loss, showing quick convergence in the first few epochs and stability following, reaching a lower validation loss at epoch 22. On the right, training and validation accuracy rapidly increase and stabilize after epoch 20, with the maximum validation accuracy at epoch 19. This shows that DenseNet121 generalizes to unseen data with minimal overfitting.

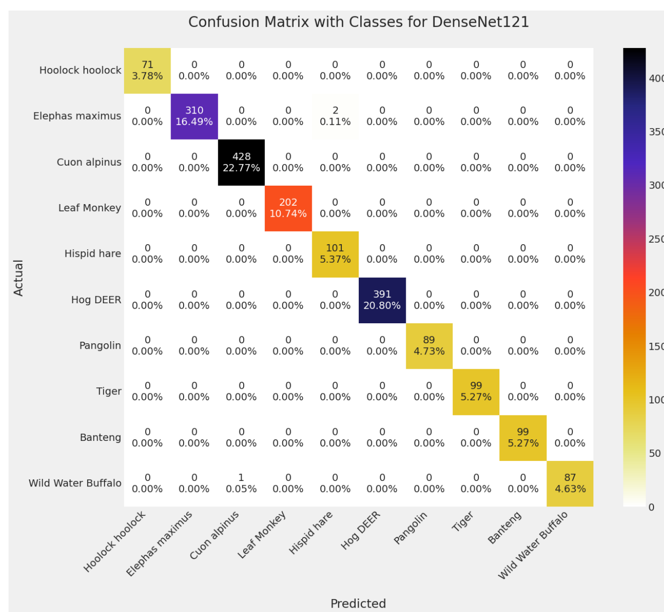

Fig. 5: Confusion Matrix of DenseNet121 model.

In Figure 5, DenseNet121's confusion matrix, which has high values along the diagonal, demonstrates its outstanding classification ability across all ten classes. Among closely related endangered species, DenseNet121 reliably and accurately distinguishes, as shown by these data.

B. Explainable Artificial Intelligence(XAI)

Explainable AI is a way to discuss an AI model's effects and flaws. When it comes to sensitive applications like endangered species detection and wildlife conservation, XAI is essential for increasing user trust, interpretability, and transparency in AI-driven systems. For real-world implementation, stakeholders must be able to understand the rationale behind specific predictions; DL models, like DenseNet121, typically behave as "black boxes." XAI approaches provide this transparency.

C. Local Interpretable Model-agnostic Explanations (LIME)

In this study, we used LIME to offer visual, easily-understood insights into the model's classification decision-making process. To find which parts of a picture (super-pixels) have the most impact on the model's predictions, LIME adjusts the input image and then watches how the predictions change. In our endangered species classification task, for example, LIME draws attention to important visual characteristics like patterns on the body, textures, or distinctive markings that are essential for differentiating between species. There can be no real-world deployment without faith in AI forecasts, particularly in high-stakes settings like wildlife monitoring. Consequently, including LIME and other XAI techniques improves the visibility of the system, makes it easier for humans to validate it, and builds trust among stakeholders. In Figure 6, we demonstrated the LIME image from the original image.

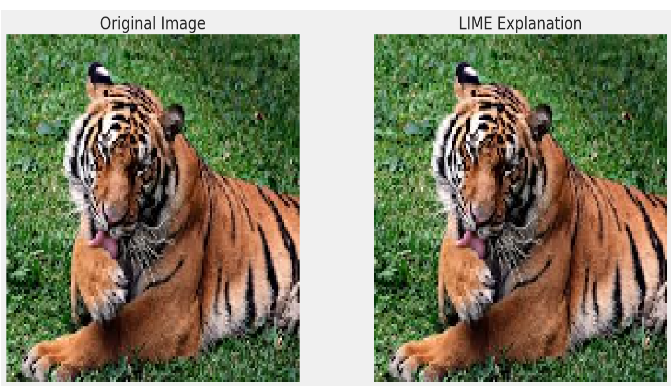

Fig. 6: LIME Explanation of the original image.

V. Conclusion and Future Work

This study presents a DL-based automated approach for identifying endangered animal species from image data. A custom dataset of endangered species was developed and subjected to standard preprocessing techniques, including resizing, normalization, and data augmentation to enhance model performance and generalization. During the training of different neural networks on endangered animal datasets, DenseNet121 had the highest classification accuracy of 99.67%. The proposed approach provides a reliable and efficient solution for endangered species monitoring, with potential applications in real-world scenarios such as preventing wildlife poaching, reducing human-animal conflicts, and minimizing animal-vehicle collisions. Additionally, Explainable AI techniques like LIME were applied to improve the interpretability of the system, offering insights into the model's decision-making process.

In our future endeavors, we aim to improve the system by incorporating real-time alert features, such as the ability to contact forest authorities when endangered animals are found. As well, enhancing the system to enable real-time video-based surveillance can bolster conservation initiatives for wildlife and assist in reducing human-wildlife conflicts.

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